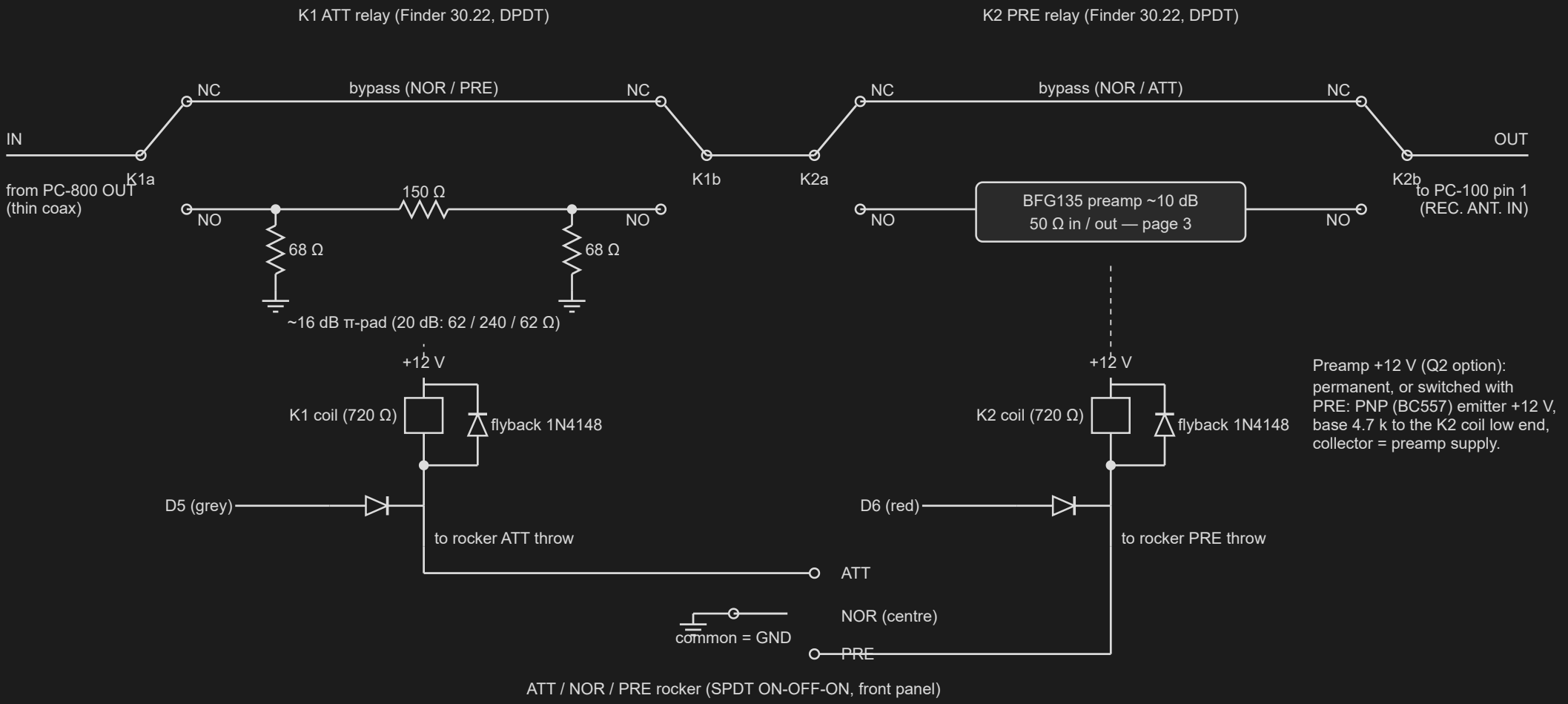
# ATT / PRE board — relays, ~16 dB attenuator, BFG135 preamp

YT3CT · 2026-10 · proposal · RX line between PC-800 OUT and PC-100 pin 1 (after the preselector). RX only — never in the TX path.




- Relays at rest (both coils off) = **NOR**: signal goes IN → K1 bypass → K2 bypass → OUT.
- ATT: K1 pulls, both poles move to NO → signal through the π-pad. PRE: K2 pulls → signal through the preamp. The rocker can only select one.
- Coil high ends permanently +12 V; the rocker grounds the coil low end. Flyback 1N4148 across each coil (band to +12 V). Sense 1N4148 from the coil low end to D5 / D6 (band to the coil) — Nano reads LOW = on.
- Thin coax (RG-174 / RG-316) in and out, short connections on the board, ground plane; keep the bypass track short and away from the preamp output.
- Check before cutting: the PC-800 → PC-100 coax must carry receive only (no TX signal).

# ATT / PRE board — PCB design notes

Everything the board needs, for drawing the PCB. Values as on pages 1 and 3.

## Connectors (board edge)

| Ref | Signal | Goes to |
|---|---|---|
| J1 | RF IN + shield | coax from PC-800 OUT |
| J2 | RF OUT + shield | coax to PC-100 pin 1 |
| J3-1 | +12 V | radio +12 V (coils, preamp) |
| J3-2 | GND | chassis |
| J3-3 | K1 coil low end | rocker ATT throw |
| J3-4 | K2 coil low end | rocker PRE throw |
| J3-5 | ATT sense | Nano D5 (grey) |
| J3-6 | PRE sense | Nano D6 (red) |

## Operation

| Rocker | K1 | K2 | Signal path | D5 / D6 |
|---|---|---|---|---|
| NOR | off | off | bypass – bypass | HIGH / HIGH |
| ATT | on | off | π-pad (−16 dB) – bypass | LOW / HIGH |
| PRE | off | on | bypass – preamp (+10 dB) | HIGH / LOW |

## Parts list (whole board)

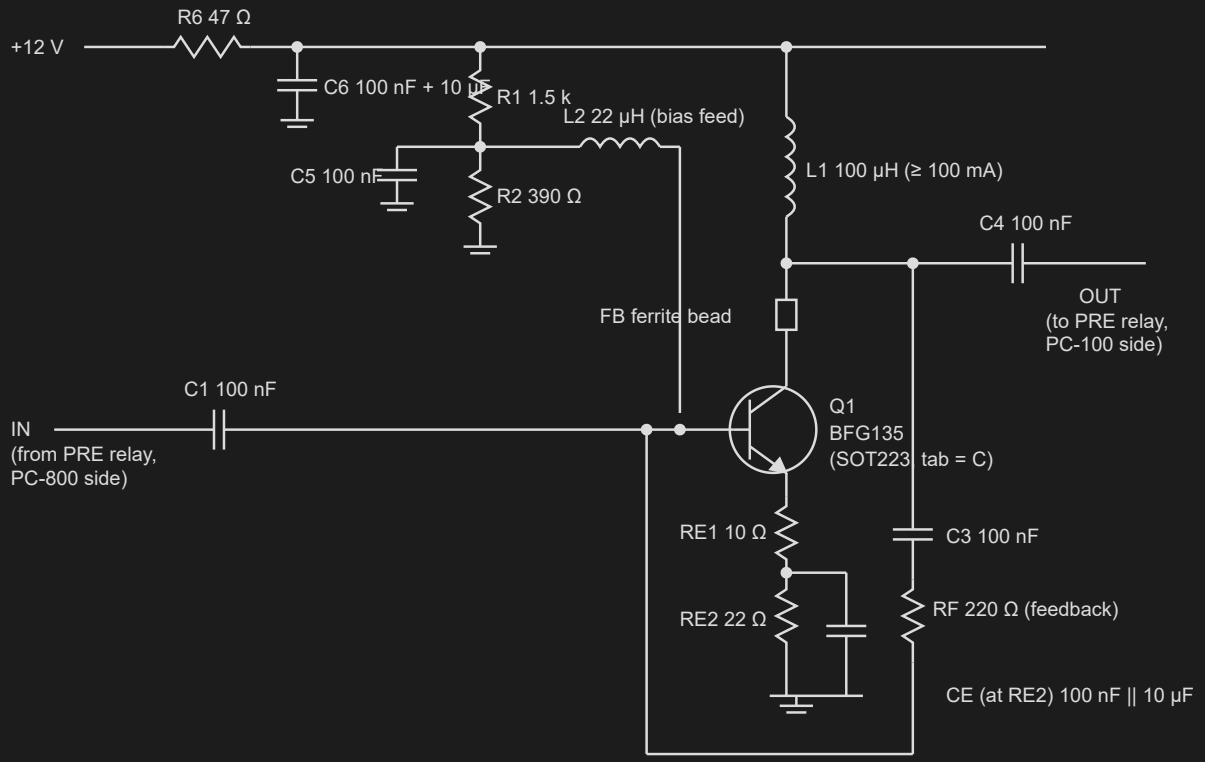
| Ref | Part | Note |
|---|---|---|
| K1, K2 | Finder 30.22.7.012 DPDT 12 V | 720 Ω coil, AgNi+Au, 5 mm pitch |
| D1, D2 | 1N4148 | flyback across K1 / K2 (band to +12 V) |
| D3, D4 | 1N4148 | sense: anode to J3-5 / J3-6, band to the coil low end |
| R10, R12 | 68 Ω | π-pad shunt (20 dB version: 62 Ω) |
| R11 | 150 Ω | π-pad series (20 dB version: 240 Ω) — 15.9 dB, $Z_{in}$ 49.3 Ω |
| Q1 + preamp | BFG135 + parts of page 3 | R1 1.5 k, R2 390, RF 220, RE1 10, RE2 22, R6 47, L1 100 µH, L2 22 µH, C1–C6, CE, ferrite bead |
| Q2 (option) | BC557 + R13 4.7 k | switches the preamp supply with PRE |
| C7 | 100 nF + 10 µF | +12 V decoupling at J3 |

## Layout

- Double-sided, bottom = solid ground plane with many vias. RF tracks short and wide.
- Signal flow in a line: J1 → K1 → K2 → J2; pad between the K1 poles, preamp between the K2 poles.
- Preamp input and output apart (no loop); BFG135 tab on a copper area; ferrite bead right at the collector.
- Coils, sense diodes and supply in a corner near J3, away from the RF path.
- Coax shields to the ground plane right at J1 / J2.

# RX preamp — BFG135 broadband feedback amplifier

YT3CT · 2026-10 · proposal · ~10–11 dB, 50 Ω in / out, 3.5–30 MHz, $I_C$ ≈ 50 mA at 12 V · sits on the PRE relay between PC-800 OUT and PC-100 pin 1




## Parts

| Part | Value | Note |
|---|---|---|
| Q1 | BFG135 | SOT223, collector tab soldered to a copper pad (heat, ~0.4 W) |
| RF | 220 Ω | sets gain / 50 Ω match (RF = Z0 · (1 + gain)) |
| RE1 | 10 Ω | unbypassed, RF feedback (RE1 = $Z0^2$ / RF) |
| RE2 / CE | 22 Ω / 100 nF ‖ 10 µF | DC stabilisation, bypassed for RF |
| R1 / R2 | 1.5 k / 390 Ω | base ≈ 2.4 V → $I_C$ ≈ 50 mA |
| L1 | 100 µH | collector choke, rated ≥ 100 mA |
| L2 | 22 µH | keeps the divider off the input |
| R6 / C6 | 47 Ω / 100 nF + 10 µF | supply filter |
| C1, C3, C4, C5 | 100 nF | ceramic, short leads |
| FB | ferrite bead | on the collector lead — against VHF oscillation |

## Notes

- Expected: **~10–11 dB** gain, 50 Ω in / out, flat 3.5–30 MHz. More gain: RF 270 Ω + RE1 8.2 Ω (~12 dB).
- Check after building: collector ≈ 9.6 V, emitter ≈ 1.6 V (= 50 mA). Adjust R2 if far off.
- BFG135 has $f_T$ ≈ 7 GHz: build on a ground-plane board, very short leads, all ground returns to the plane.
- Supply: permanent +12 V, or switched on with PRE (PNP from the PRE coil low end) to save ~60 mA.
- On the PRE relay: pole 1 switches the input, pole 2 the output (bypass when off). RX only — never in the TX path.
- Test with the tinySA (sweep through the amp, 50 Ω both ends) or by ear on 10 / 15 m.